

Justin Lee

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EDUCATION

San Jose State University

Bachelor of Science in Computer Science - 3.73/4.0

San Jose, CA

Dec. 2025

SKILLS

Languages: Javascript, Python, SQL, Java

Front-end: React.js, Next.js, Tailwind CSS

Tools: AWS, Docker, GitHub Actions (CI/CD), GCP

Back-end: Express.js, Django, PostgreSQL, MongoDB

PROFESSIONAL EXPERIENCE

Software Engineer Intern

Delta Labs AI

Nashville, TN

May. 2025 - Present

- Built frontend of MVP using React.js, TypeScript, and TailwindCSS, delivering 7+ responsive pages with modular architecture ensuring clean code organization, component reusability, and long-term scalability.
- Refactored a legacy Flask backend by introducing a DAO layer and implementing connection pooling using psycopg2, significantly improving code modularity and reducing average query latency by over 70%.
- Restructured and designed normalized PostgreSQL schema with 20+ tables and integrated role-based access control.
- Deployed full-stack app on Google Cloud Compute Engine with Dockerized NGINX, React, and Flask services, using managed infrastructure like Cloud SQL and GCP Secrets Manager for secure, scalable deployment.

Software Engineer Intern

Cadence Design Systems

San Jose, CA

Jul. 2025 - Aug. 2025

- Rebuilt a 5-year-old HTML-to-PDF generator for the Technical Publications team, designing a modern Python solution with BeautifulSoup to combine and standardize 70+ HTML files into a single, professional PDF document.
- Proposed and implemented a scalable architecture by converting individual HTML files into PDFs before merging, enabling processing of 1,000,000+ lines of HTML without crashing headless browser tools like Playwright.
- Engineered a dynamic table of contents system using PyMuPDF to locate and link bounding boxes of headings in the output PDF, ensuring clickable TOC links remained accurate across merged documents.
- Optimized script runtime by over 60% through multiprocessing and efficient memory management, reducing generation time from ~20 minutes to under 8 minutes on large file sets.

Lead Software Engineer

Arctex

Corvallis, OR

Sep. 2024 - Aug. 2025

- Dockerized and deployed CollegeAppAssist, a MERN app using AWS Fargate and ALB for scalable hosting, resulting in stable performance and reliably serving 25,000+ daily requests with Cloudflare-enabled protection.
- Built a CI/CD pipeline using GitHub Actions to automate deploying to AWS Fargate, eliminating the need for manual intervention and saving 5+ minutes per deployment, accelerating development iteration and reducing friction.
- Implemented session-based authentication with OAuth, enabling secure user login for 300+ monthly users.
- Integrated infrastructure for payment workflows, including Stripe Webhooks to handle real-time transactions.
- Designed a flexible MongoDB schema to efficiently store large amounts of variable HTML ID fields across multiple websites, supporting over 100 top-level attributes and deeply nested structures for consistent and scalable data querying.

PROJECTS

ACM-CS @ SJSU Lenses

Jan. 2024 - May. 2025

Tech Stack: Django, PostgreSQL, Next.js, Docker, OCI

- Scraped and cleaned 25,000+ raw data entries consisting of reviews, professor data, course information, and semester specific schedules to build a reliable dataset for an SJSU Course Scheduler Application.
- Oversaw the development of a relational database using PostgreSQL that holds 12 tables and 30,000+ data entries.
- Developed a secure login system with Google OAuth, implementing token management and automatic refreshing.
- Designed and refactored 20+ RESTful API endpoints for CRUD operations using Django to replace inefficient data handling logic, reducing backend response times and improving maintainability across services.

TheTypingSchool

Jun. 2024 - May. 2025

Tech Stack: Express.js, EJS, OpenAI API, MongoDB, AWS

- Developed a leveling and rewards system for 50+ monthly users, boosting engagement and progress.
- Refactored the Express.js codebase for modularity and performance, reducing endpoint response times by 20%.
- Integrated OpenAI's Batch API with node-cron to auto-generate 100+ monthly typing prompts, replacing manual content creation and enhancing challenge variety for increased user retention.
- Built an eye-tracking feature using WebGazer.js with a calibration screen to block typing when users looked down, reducing gaze-down events by 30%, improving learning outcomes and focus during typing challenges.